



Laboratory Practice 02

Polymerase Chain Reaction (PCR)

INTRODUCTION

Polymerase Chain Reaction (PCR) is an in vitro exponential amplification reaction of a specific DNA sequence. The reaction requires: small amounts of DNA (even a single copy of the sequence to be amplified); specific oligonucleotides known as primers, that complementarily bind to flanking sequences of the DNA fragment to be synthesized, deoxynucleotides (dNTPs) as well as DNA polymerase (Taq polymerase).

The reaction of DNA amplification consists of three stages: denaturation, hybridization and extension; these are repeated for several amplification cycles to synthesize new DNA molecules. After a certain number of cycles, the amount of DNA is such that it can be visualized by agarose gel electrophoresis, stained with GelRed.

PCR has multiple applications, including direct DNA cloning, site-directed mutagenesis, genetic engineering, fingerprinting, detection of infectious agents, allelic variant analysis, DNA sequencing, among others.

OBJECTIVES

- Determine the presence of a DNA insert in the recombinant plasmid from an *E. coli* culture.
- Identify the amplicon's size amplified from the cloned gene.

MATERIALS AND EQUIPMENT

- Thermal cycler.
- Micropipettes 2 - 20 μ L, 20 - 200 μ L.
- Autoclaved tips.
- 1.5 mL microcentrifuge tubes.
- 0.2 mL tubes.
- Taq DNA Polymerase Buffer 10X.
- MgCl₂ 25 mM.
- dNTP Mix 2.5 mM.
- Taq polymerase (0.5 U/ μ L).
- Primer Set Forward (pGLOF) and Reverse (pGLOR).
- Sample: *E. coli* culture with "pGLO".
- Positive control: purified DNA insert.
- Negative control: PCR grade H₂O.
- Microcentrifuge tube racks.
- Agarose gel 1.2 %.
- TBE Buffer 0.5X.
- Horizontal electrophoresis chambers.

METHODOLOGY

1. Take a 1.5 mL microcentrifuge tube, add 40 μL of H_2O PCR, then add 10 μL of *E. coli* culture. Place at 95 $^{\circ}\text{C}$ for 5 min. Meanwhile, work on step 4.
2. Centrifuge at 13,000 rpm for 1 min at room temperature.
3. Recover the supernatant and place it in a new tube.
4. Place three 0.2 mL tubes in a rack. Label the tubes and keep them in the same position on the rack. The order of the tubes shall be as follows:
 - **Tube 1:** Negative Control
 - **Tube 2:** “pGLO” from culture’s supernatant
 - **Tube 3:** “DNA” purified
5. A Master Mix has already been prepared, as the volumes to be used are quite small and pipetting errors needed to be avoided, see the table below:

Table 1: Master Mix for PCR

Components	Volume	
	1 Rx (μL)	3 Rx (μL)
10X Buffer	1.0	3
MgCl_2 (25 mM)	0.8	2.4
dNTPs (2.5 mM c/u)	0.8	2.4
Primer Forward (20 μM)	0.25	0.75
Primer Reverse (20 μM)	0.25	0.75
Taq Polymerase (5 U/ μL)	0.04	0.12
H_2O PCR	1.86	5.58
Total (μL)	5	15

6. Place 5 μL of the master mix per reaction tube and add the DNA according to Table 2.

Table 2: Distribution of volumes per reaction tube

	Negative control (μL)	Supernatant (μL)	Purified DNA (μL)
Master Mix	5	5	5
H_2O PCR	5	0	3
Supernatant	0	5	0
DNA purified	0	0	2

7. Cover the tubes and place them in correlative order and in the designated positions according to the teacher’s instructions. Remember these tubes must be closed and labeled.

8. PCR conditions will be programmed in the thermal cycler as detailed in Figure 1:

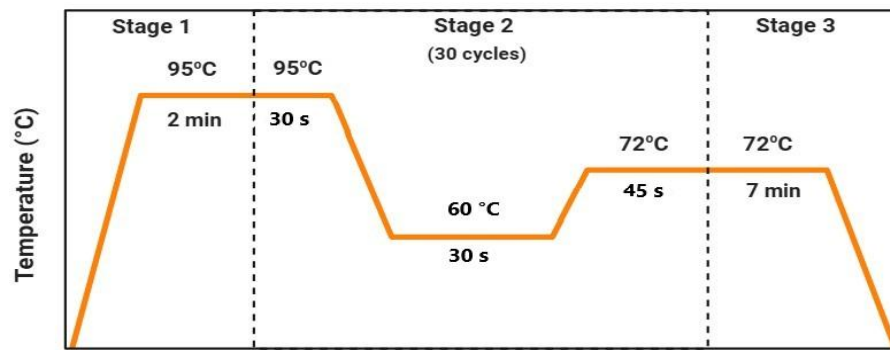


Figure 1: DNA amplification conditions to be programmed in the thermal cycler. The hybridization temperature of the pair of primers pGLOF and pGLOR is 60 °C.

9. Upon completion of the PCR reaction, carefully remove the tubes from the thermal cycler, and place them on ice.
10. Approach to the electrophoresis chamber with your samples and material, there, mix 10 µL of your product with 2 µL of 6X DNA loading buffer and analyze the PCR products by running an electrophoresis in 1.2 % agarose gel in TBE buffer 0.5X.
11. A picture from the agarose gel will be uploaded to the Blackboard platform. Determine the size of the amplification products (amplicons).

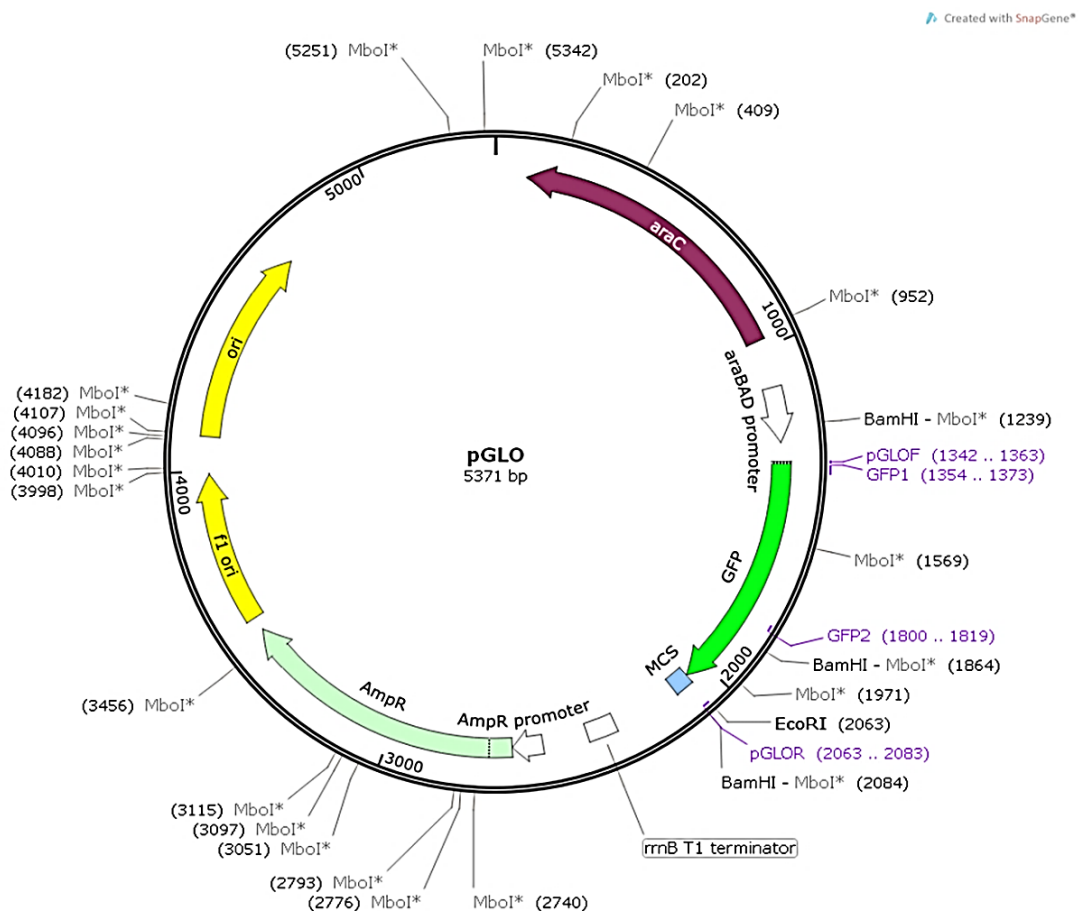


Figure 2: pGLO plasmid, showing hybridization regions for primers pGLOF & pGLOR

QUESTIONNAIRE

1. What are the main phases of a PCR reaction? Describe them.
2. What is the role of Taq polymerase in PCR?
3. What is the final concentration of primers, MgCl_2 , dNTP's, and Taq polymerase units used in each reaction?
4. Figure 2 shows the plasmid pGLO, the arrows indicate the hybridization position of the Forward and Reverse primers. Calculate the size of the amplification product.
5. You have received a set of lyophilized primers, the concentration of each is 40 nmol each. Determine the amount of PCR water to use to resuspend the primers at a concentration of 200 μM .

REFERENCES

- Ausubel F., Brent, Kingston R., Moore D., Seidman J., Smith J. y Struhl K. 1989. Current Protocol in Molecular Biology, Greene Publishing Associates and Wiley-Interscience.
- Sambrook, J & D, Russell. 2001. Molecular cloning: Laboratory Manual. Vol 1. Editorial CSHL. EE.UU.